Project Title

Local Governments Survey

Community Data Lab Project Report:

ARISE

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Date – 6 May 2026

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SUMMARY

In order to produce a single dataset for evaluating local government capacity in Kansas, this project focuses on harmonizing two separate survey datasets: the CivicPulse Kansas Sample and the ARISE Kansas City Survey. The concept of local government capacity includes infrastructural resilience, financial stability, and workforce availability. Since both datasets differ in structure, wording, and response scales, direct comparison requires a systematic harmonization process.

Data cleaning, codebook interpretation, crosswalk generation, and harmonized variable creation were used to align both datasets into a unified analytical framework. A merged dataset of 444 observations was created using data standardization, transformation, and integration techniques. Regression modeling, comparative analysis, and exploratory analysis were then applied to identify patterns and relationships among key variables.

The findings show that local governments consistently face workforce challenges, while their relationship with infrastructure stress and fiscal condition remains relatively weak. Overall, this study highlights the importance of data harmonization in transforming fragmented datasets into meaningful insights for policy and decision-making.

INTRODUCTION

Local governments are essential to the provision of public services, upkeep of infrastructure, and general community well-being. A number of interrelated elements, such as worker capacity, financial stability, and infrastructural resilience, affect their efficacy. Policymakers, scholars, and administrators who want to enhance resource allocation and governance must comprehend these elements.

In reality, independent polls with distinct objectives are frequently used to gather information about local government capacity. These surveys often differ in terms of organization, vocabulary, and response forms, even though they may measure comparable themes. Because variables cannot be directly aligned without rigorous interpretation and transformation, this poses a serious problem for comparative analysis. Important insights are therefore still dispersed throughout datasets. The ARISE Kansas City Survey and the CivicPulse Kansas Sample, two survey datasets that capture important facets of local government operations like workforce issues, financial conditions, and infrastructure-related problems, are integrated in this project to address this difficulty.

By aligning overlapping constructs and standardizing variables into a uniform analytical framework, this effort aims to provide a harmonized dataset. Codebook interpretation, crosswalk variable mapping, and data transformation into similar formats are all part of this process. Beyond technical integration, the project focuses on comprehending how various aspects of local government capability interact and can be examined together.

The goal of this project, which was carried out in cooperation with ARISE as the community partner and sponsor, was to integrate the findings from the two surveys in

order to gain a deeper understanding of Kansas local government capacity. The study concentrated on aligning variables and producing a single dataset that allows for meaningful comparison across sources because the surveys cover similar subjects but differ in question structure and measurement. The harmonization process included variable recoding, proxy construction, standardization, and composite index development.

Along with a written report that examines important topics like workforce difficulties, the state of the economy, and infrastructure-related problems, the initiative provides a harmonized dataset. The analysis enables a better understanding of capacity limits and offers significant insights into the difficulties faced by local governments by combining both datasets into a similar analytical framework. These insights can support more efficient planning and resource allocation as well as help guide policy decisions.

PREPARATION

The project's preparation phase is essential because it lays the groundwork for precise analysis and insightful interpretation. The CivicPulse Kansas Sample and the ARISE Kansas City Survey have different response formats, variable names, and structures because they were gathered separately. Due to these variations, direct comparison is difficult and a methodical methodology is needed to guarantee consistency across datasets.

Understanding the datasets using their corresponding codebooks was a crucial part of this phase. The codebooks provide crucial details on response scales, coding systems, and variable definitions. Without this context, numerical values could be misconstrued since,

depending on the type of question, the same figures could have multiple meanings. Therefore, before doing any data transformation or cleaning, the codebooks have to be carefully interpreted.

Understanding Dataset Structure :

The ARISE Kansas City Survey and the CivicPulse Kansas Sample are two independently gathered survey datasets used in the research that capture various facets of local government capacity. The ARISE dataset comprises a range of variable types, including binary indicators, ranking-based variables, Likert-scale responses, and comprehensive questions about infrastructure systems, governance, fiscal condition, and hazard preparedness. For instance, ordinal scales with a range of 1 to 5 are used to measure issues with the workforce and the financial situation. The CivicPulse dataset, on the other hand, has a separate name scheme and coding structure but concentrates on workforce, financial, and infrastructure-related challenges.

Although these datasets cover similar themes, direct comparison is difficult due to differences in response coding, variable naming, and structure. To find comparable variables and guarantee consistency across datasets, these discrepancies needed to be carefully examined. In order to accurately comprehend the meaning of variables and answer scales, codebook interpretation was also necessary because multiple concepts may be represented by the same numerical values depending on the context of the question. This demonstrated the necessity of using a methodical harmonization technique prior to conducting analysis.

Data Cleaning and Standardization:

To ensure both datasets were analysis-ready, a structured data cleaning process was applied. Blank entries and inconsistent labels like "NA" and "Don't know" were transformed into a consistent format in order to standardize missing values. To prevent misunderstandings during analysis, null values were also substituted for numerical placeholders (such as 99 or -99).

Additionally, data types were aligned between the two datasets and column names were standardized into a uniform format. Formatting discrepancies were eliminated by cleaning text-based variables. Before merging, data validation checks were carried out to confirm row counts (309 ARISE and 135 CivicPulse), validate dataset structure, and guarantee correct column alignment. These actions enhanced the reliability, consistency, and transparency of the data.

Crosswalk Development and Harmonization Framework:

A data harmonization procedure was carried out to allow for a meaningful comparison between the ARISE and CivicPulse datasets. There were differences in the variable names, formats, and measurement structures between the two datasets since they were gathered using different survey instruments. A crosswalk table was created to consistently align conceptually related variables under standardized names in order to address this.

Every variable pair was thoroughly examined during this procedure and categorized as a derived variable, no match, weak partial match, precise match, or partial match. In order to prevent inaccurate comparisons between unrelated variables, this structured

classification made sure that only significant and theoretically sound associations were kept. Consequently, the crosswalk enabled consistent, dependable integration of both datasets and functioned as the basis for all subsequent harmonization procedures.

Three primary methods - direct matching, conceptual alignment, and proxy construction were used to steer the harmonization process. When variables in both datasets measured the same concept using comparable scales, direct matching was employed. For instance, since both measure workforce issues on a 1–5 scale, the variable workforce_challenge_level was established by mapping ARISE Q8 with CivicPulse workforce_sc. Similarly, ARISE Q119 and CivicPulse condition_sc, which both use similar rating scales to reflect overall fiscal condition, were used to align fiscal_condition_overall.

Conceptual alignment was used when variables reflected comparable underlying concepts but were not identical. For example, similar metrics from various datasets that capture the financial strain on local governments were combined to create infrastructure financing stress. There was no direct variable for infrastructure stress in the ARISE dataset, especially from Q14. Related factors from Q13 were utilized as proxies to address this, including decreased services, erratic funding, and postponed capital projects. Despite their different wording, these variables measure similar financial challenges. To ensure comparability with CivicPulse infrastructure-related metrics, they were merged and standardized to provide a composite infrastructure_finance_stress_index. Before creating the final infrastructure_finance_stress_index, z-score normalization was applied within each dataset because the ARISE and CivicPulse infrastructure variables were measured differently.

Similarly, CivicPulse governance indicators, which represent planning, stakeholder cooperation, and organizational practices, were combined with ARISE Q7, which assesses elected official support, to harmonize governance-related metrics. A single governance_practice_index was used to group these variables.

A new strategy was needed for variables relating to services. While CivicPulse employed category indications, ARISE already tracked service availability in binary form (chosen/not selected). All service variables were recoded into binary format, with 1 denoting service availability and 0 denoting absence, in order to standardize them. This made it possible to compare the two datasets consistently.

Additionally, a contextual alignment approach was used to harmonize variables pertaining to interorganizational interactions. The CivicPulse partner_sc variable, which records nonprofit and interagency collaboration, was matched with ARISE Q12, which gauges collaboration problems. Both variables show similar aspects of organizational coordination, notwithstanding their differences.

All things considered, this harmonization approach produced a collection of standardized variables that encompass important aspects including workforce difficulties, financial health, infrastructure stress, grant capacity, governance procedures, and service accessibility. While maintaining the fundamental meaning of each construct, this alignment allowed for integrated analysis across both datasets.

Creation of Harmonized Variable Columns:

Nineteen common variable columns were found and harmonized between the two datasets based on the crosswalk. These factors, which include workforce challenge level,

fiscal condition, infrastructure finance stress, preparation indicators, and support for vulnerable people, are important aspects of local government capacity.

Variables were converted into a similar scale, usually between 1 and 5, to guarantee comparability. Composite indices were constructed utilizing several related variables when direct matches were unavailable. To guarantee consistency across datasets, binary service variables were recoded into 0/1 indications. To ensure consistency in the analysis, interpretations were standardized for both datasets.

Final Dataset Preparation:

Both datasets were pooled into a single consolidated dataset following harmonization. To guarantee consistency and clarity, only pertinent, aligned variables were kept, and a source column was added to differentiate between ARISE and CivicPulse observations.

There are Nineteen important variables and 444 total records in the final dataset. This dataset offers a solid basis for examining trends and relationships related to local government capacity, as it is well-structured, consistent, and prepared for analysis. Contextual identifiers including source dataset, government type, and jurisdiction information, when accessible, were also included in the final dataset.

ANALYSIS

In order to gain a better understanding of local government capacity in Kansas, the analytical phase concentrated on looking at patterns, distributions, and linkages within the harmonized dataset. Descriptive statistics, distribution analysis, cross-tabulations, regression modeling, correlation analysis, and comparison analysis across datasets were

among the analytical methods used with the merged dataset. These techniques provide both broad and in-depth insights into important factors like personnel difficulties, financial health, and infrastructural stress.

Finding patterns, comparing responses from the ARISE and CivicPulse datasets, and assessing the connections between key factors were the objectives of the investigation. Bar charts, histograms, boxplots, and stacked bar charts are examples of visualizations that were employed to enhance interpretability and bolster conclusions. For ARISE and CivicPulse, regression models were generated independently before being integrated using a source-control model.

Additionally, statistical methods such cross-tabulations, and correlation analysis were used to evaluate the consistency and strength of correlations between datasets. When combined, these methods allowed for a thorough knowledge of the ways in which infrastructure, workforce, and financial issues interact and impact local government capability.

Descriptive Analysis:

To better understand variations in local government capability, significant variables from the ARISE and CivicPulse datasets were compared using visualizations. The average values of workforce difficulties and financial situation were examined using bar charts, histograms and box plots which made it easy to compare different datasets. These graphics aid in identifying trends, bolstering statistical conclusions, and enhancing the analysis's readability.

Figure 1

Figure 1 uses a count plot to show the distribution of workforce challenge levels across all data. Workforce challenge levels are represented on a scale from 1 (low) to 5 (high) on the x-axis, and the number of responses is displayed on the y-axis. Most local governments face moderate to high workforce issues, as evidenced by the concentration of replies at levels 3 and 4, each with over 130 observations. Level 1 contains the fewest responses about 15 while Levels 2 and 5 exhibit moderate frequency of over 60 responses apiece. With comparatively few respondents indicating low levels of difficulty, this distribution indicates that workforce issues are widespread throughout the dataset rather than uncommon. Workforce capacity is a crucial component of local government performance in the context of this study, and this pattern shows that staffing and workforce-related difficulties are a persistent concern that may have an impact on the organization's overall effectiveness, service delivery, and financial situation.

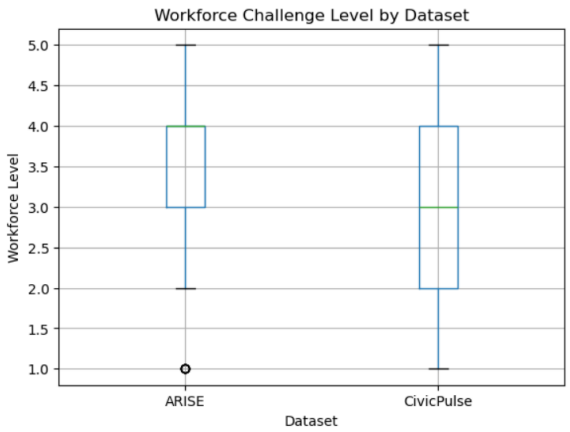

Figure 2

A boxplot comparing the workforce challenge levels from the ARISE and CivicPulse datasets is shown in Figure 2. On a scale of 1 (low) to 5 (high), the boxplot shows the distribution, median, and variability of responses for each dataset. The majority of responses in the ARISE dataset fall between 3 and 4, indicating comparatively higher and more consistent workforce issues. The median workforce challenge level is approximately 4. The CivicPulse dataset, on the other hand, shows more diversity in reported problems, with a median of roughly 3 and a broader spread between roughly 2 and 4.

ARISE responses are somewhat more focused with fewer low-end numbers, albeit there is one low outlier close to level 1, whereas CivicPulse displays a wider overall range, spanning from 1 to 5. This comparison shows that although both datasets show moderate to high workforce issues, CivicPulse captures more variation across local governments, while ARISE respondents often report slightly higher and more consistent levels. This change in the study's setting draws attention to variations in workforce capacity circumstances across datasets and underscores the significance of workforce issues as a major factor affecting local government performance.

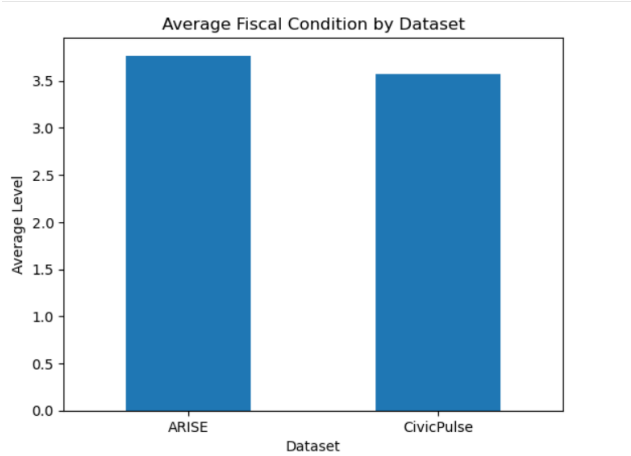

Figure3

The average fiscal situation for the ARISE and CivicPulse datasets is displayed in Figure 3, with the ARISE dataset having a mean of roughly 3.77 and the CivicPulse dataset having a slightly lower mean of about 3.57. On a scale of 1 to 5, both scores are in the mid-to-high range, suggesting that local governments in both datasets usually view their financial situation as stable. Despite variations in respondents and structure, there appears to be significant consistency across the two datasets, as indicated by the modest discrepancy of roughly 0.2 points. This is crucial for the project because it shows that financial conditions are a dependable and comparable variable across datasets, supporting cross-dataset analysis and confirming the conclusion that financial conditions are less variable and more stable than other factors like infrastructure stress or workforce challenges.

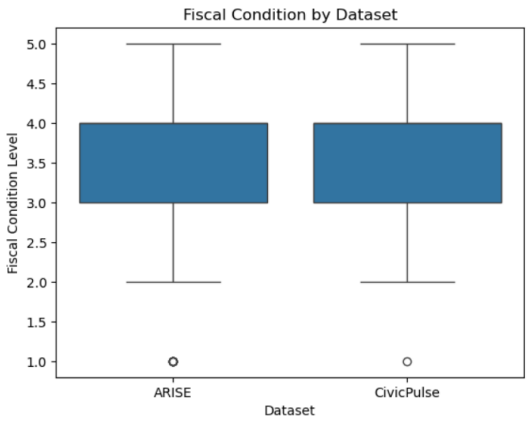

Figure 4

A boxplot comparison of the fiscal conditions of the ARISE and CivicPulse datasets is shown in Figure 4. The majority of local governments report solid financial situations, as seen by the median fiscal state of roughly 4 in both datasets. The middle 50% of responses are closely grouped and consistent, as indicated by the interquartile range (IQR) for both datasets, which is approximately between 3 and 4. With a few lower-value outliers around 1, the overall range spans from roughly 2 to 5, indicating that only a limited percentage of jurisdictions have financial difficulties. Fiscal conditions are very constant between ARISE and CivicPulse, as evidenced by the similarity in medians, dispersion, and distribution between the two datasets. This supports the project's conclusion that the financial situation is steady and comparable.

Fiscal conditions are very constant between ARISE and CivicPulse, as evidenced by the similarity in medians, dispersion, and distribution between the two datasets. This supports the project's conclusion that, in comparison to other variables like workforce problems or infrastructure stress, fiscal situation is a stable and similar characteristic that exhibits less variability.

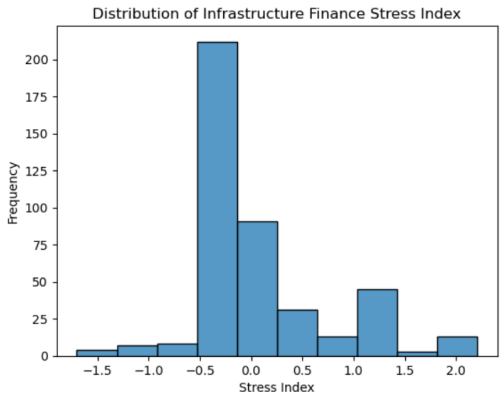

Figure 5

The infrastructure finance stress index's distribution throughout the dataset is displayed in Figure 5. The majority of data fall between -0.5 and 0.5, with a mean value of roughly 0.00, indicating usually moderate levels of financial stress associated to infrastructure. With values ranging from roughly -1.6 to 2.2, the distribution is somewhat dispersed, indicating that some jurisdictions have noticeably lower or greater stress levels. The occurrence of higher positive values indicates that some local governments are under significant financial strain in relation to infrastructure, and the standard deviation of roughly 0.72 indicates moderate variability. This data supports the general conclusion that infrastructure-related issues vary greatly among jurisdictions by highlighting how infrastructure stress is more varied and unevenly distributed within the project's context than budgetary situation.

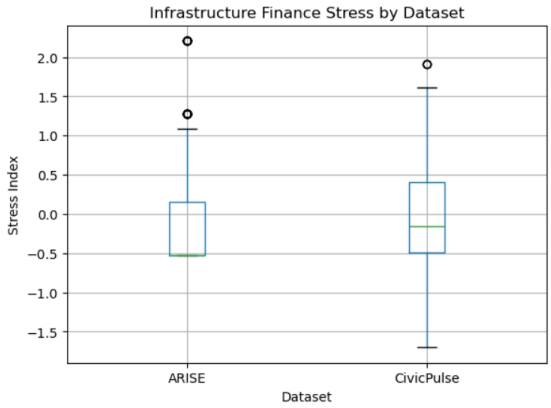

Figure 6

A boxplot comparison of the infrastructure financing stress index between the ARISE and CivicPulse datasets is shown in Figure 6. Both datasets show moderate levels of infrastructure-related financial stress, with ARISE's median value being roughly -0.5 and CivicPulse's significantly higher median being around -0.2. ARISE's interquartile range (IQR) is about between -0.6 and 0.2, while CivicPulse's is larger, roughly between -0.5 and 0.4, indicating that the CivicPulse dataset has more variability.

Additionally, CivicPulse's general range is wider (approximately -1.7 to 1.9) than ARISE's (about -0.7 to 2.2), and both datasets contain positive outliers above 1.0, suggesting that certain jurisdictions have noticeably greater stress levels. This graphic supports the conclusion that infrastructure stress is less consistent and more dataset-dependent than stable variables like fiscal situation in the context of the project by highlighting how uneven and variable it is across datasets.

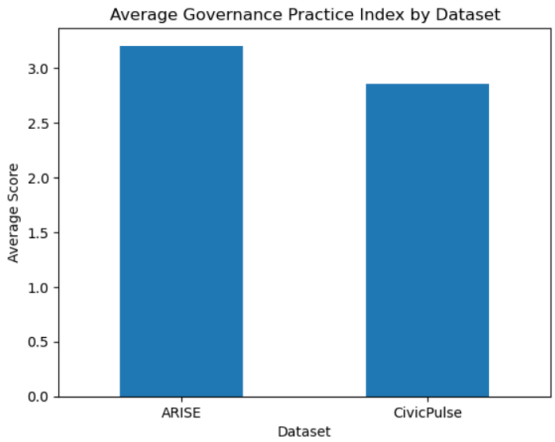

Figure 7

A bar graph comparing the average governance practice index between the ARISE and CivicPulse datasets is shown in Figure 7. ARISE respondents indicate somewhat better governance processes, as evidenced by their mean score of roughly 3.2 compared to CivicPulse's somewhat lower average of about 2.9. The little difference, however, indicates that governance practices and institutions are largely the same in both datasets. This image illustrates how harmonized variables allow for meaningful comparison of institutional practices across datasets and supports the notion that governance-related capacity is relatively stable across jurisdictions in the context of the project.

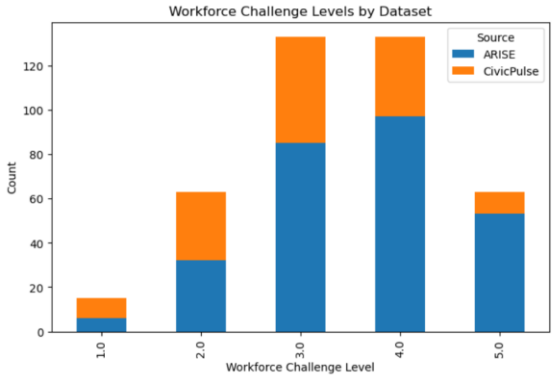

Figure 8

Figure 8 uses a stacked bar chart to display the distribution of workforce problem levels across the ARISE and CivicPulse datasets. The bulk of responses in both datasets are centered at levels 3 and 4, suggesting that local governments typically face moderate to severe workforce issues. In comparison to CivicPulse, the ARISE dataset typically has higher counts in these areas, indicating marginally higher perceived workforce problems. By displaying the distribution of responses across categories, this visualization offers a more comprehensive view than average-based charts. It highlights the disparities in response patterns between the two datasets and supports the project's main finding that workforce issues are pervasive and consistently present across jurisdictions.

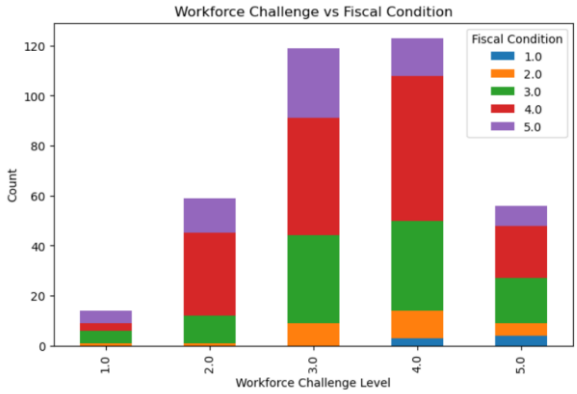

Figure 9

The association between workforce problem levels and fiscal situation is depicted in Figure 9 using a stacked bar chart. The majority of jurisdictions express moderate to high workforce issues, as seen by the greatest counts at workforce challenge levels 3 and 4, when total replies surpass roughly 120 observations. The most common fiscal condition ratings within these categories are 3 and 4, indicating that financial conditions are still quite steady even in areas with greater workforce issues. Levels 1 and 2 of the workforce

challenge feature substantially fewer observations (usually less than 60 responses), and level 5 likewise exhibits a decrease in overall responses. There is no discernible increasing or decreasing trend in the distribution of fiscal conditions across all worker levels.

There is no discernible increasing or decreasing trend in the distribution of fiscal conditions across all worker levels. This implies that the state of the economy is not significantly affected by workforce challenges. Overall, the statistic confirms the conclusion that there is little association between workforce difficulties and financial health.

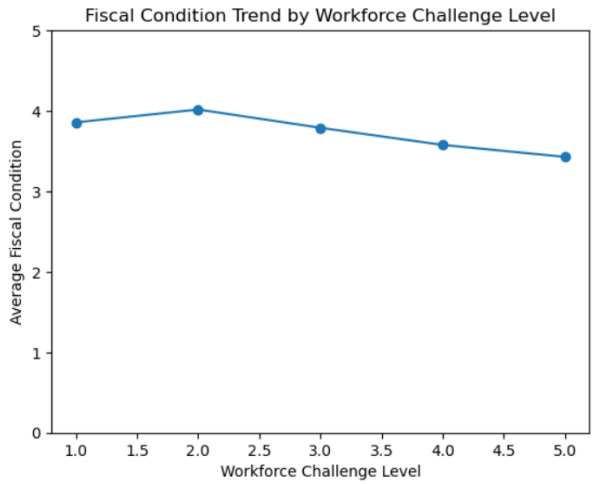

Figure 10

A line graph illustrating the correlation between average fiscal condition and workforce challenge levels is presented in Figure 10. The fiscal situation peaks at workforce level 2 (≈4.0), is somewhat worse at level 1 (≈3.8), and then progressively deteriorates as workforce problems rise, falling to roughly 3.8 at level 3, 3.6 at level 4, and 3.4 at level 5. This pattern points to a little unfavorable tendency in which slightly worse financial conditions are linked to more workforce challenges. The link is

comparatively weak, though, as the changes are not very significant. This figure reinforces the conclusion that these variables are only tangentially related, supporting the project's overall finding that, although workforce difficulties have some influence, they do not significantly predict fiscal health.

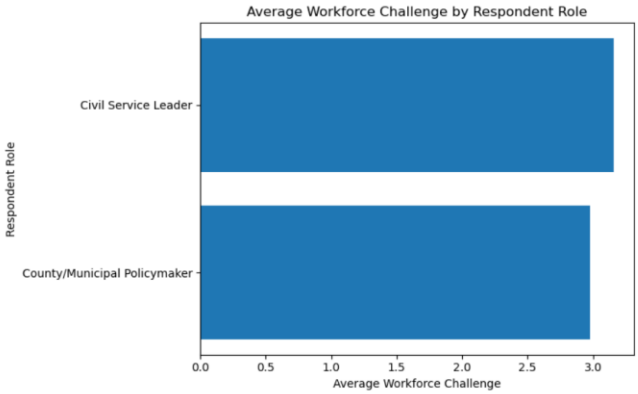

Figure 11

A horizontal bar chart comparing the average workforce difficulty levels by respondent role is shown in Figure 11. County/Municipal Policymakers report a slightly lower average of about 3.0, whilst Civil Service Leaders indicate a slightly higher average workforce challenge level of about 3.2. This suggests that both groups have a moderate understanding of workforce concerns, with administrative and policymaking viewpoints differing just slightly. In the project's context, this figure implies that workforce challenges are consistently acknowledged across various local government jobs, supporting the validity of the results and showing consistency in the experiences and reporting of workforce-related challenges.

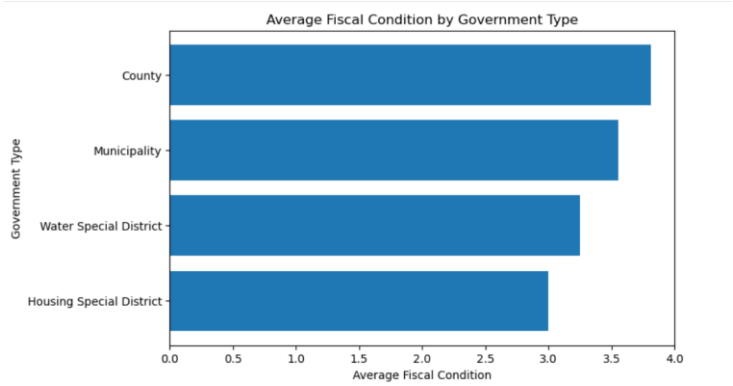

Figure 12

The average fiscal situation for various forms of government is depicted in Figure 12. At roughly 3.8, counties report the best average fiscal position, followed by municipalities at roughly 3.6. Housing special districts have the lowest average, at roughly 3.0, while water special districts have an average of 3.3. Notwithstanding these variations, the range is generally quite small (about 3.0 to 3.8), suggesting little variation among government types. Every category is in the mid-to-high range, indicating that the majority of local governments believe their financial situation is stable. The general conclusion that fiscal conditions are comparatively uniform and do not differ considerably by type of government is supported by this consistency across categories.

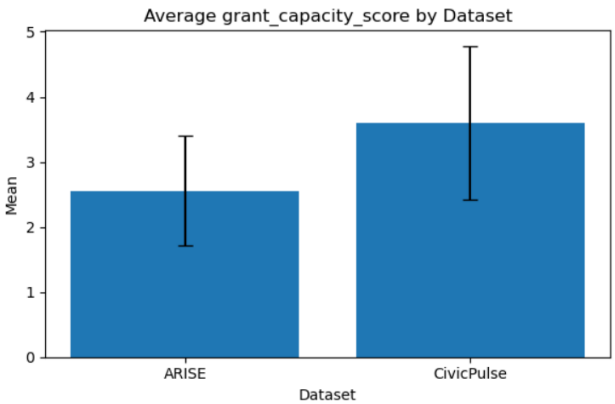

Figure 13

The average grant capacity score by dataset is displayed in Figure 13. Compared to ARISE, which has a mean score of around 2.5, the CivicPulse dataset has a higher mean score of about 3.6, suggesting that respondents in CivicPulse report a superior ability to receive and handle funds. The wider error bars indicate that CivicPulse has a higher standard deviation (about 1.2) than ARISE (about 0.9), indicating more response variability. This indicates that although CivicPulse exhibits a greater overall grant capacity, the experiences of specific jurisdictions differ more greatly. A discernible gap across the datasets is highlighted by the difference in mean values, indicating that dataset-specific factors like respondent characteristics or geographical conditions may have an impact on grant capacity. This is significant for the research because grant capacity is essential to maintaining financial stability and allowing local governments to finance services and infrastructure.

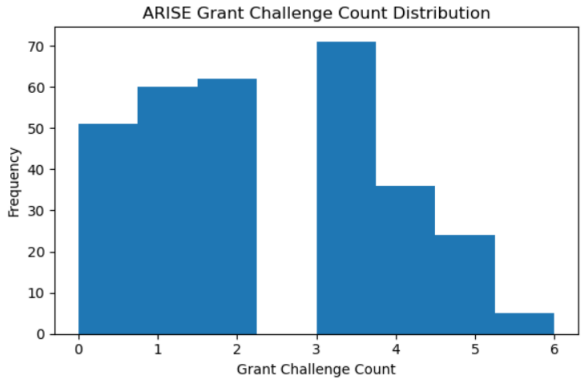

Figure 14

The distribution of grant challenge counts in the ARISE dataset is displayed as a histogram in Figure 14. While fewer local governments report very high counts, the majority of observations fall in the low to moderate range (around 1-3), suggesting that many local governments face a reasonable number of grant-related issues. This implies

that there is variation in the challenges encountered when obtaining or handling funding. This is crucial for the project since funding capacity is impacted by grant issues, which in turn affects financial stability and the capability to support services and infrastructure.

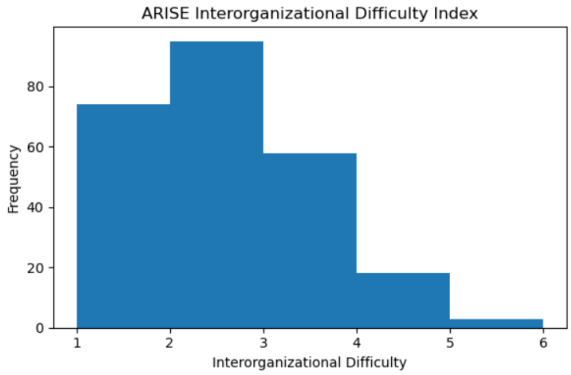

Figure 15

Local governments' challenges in working with other groups are depicted in Figure 15: Distribution of Interorganizational Difficulty (ARISE Dataset). The majority of responses are clustered between 2 and 3, with values ranging from roughly 1 to 6, suggesting that most respondents have low to moderate cooperation issues. High levels of coordination difficulties are less common since the number of observations drops as the difficulty level rises and very few cases show above 4. All things considered, this implies that although there are interorganizational difficulties, they are usually controllable. Understanding these difficulty levels aids in assessing how coordination affects local government capability, which is crucial for the project since efficient collaboration is necessary for infrastructure design, service delivery, and overall government performance.

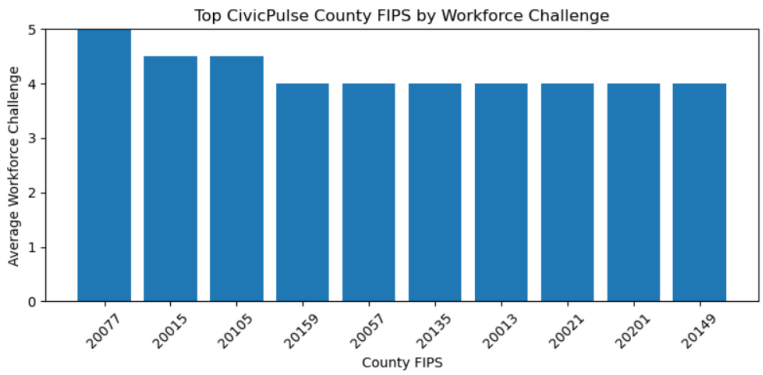

Figure 16

The top CivicPulse county FIPS (Federal Information Processing Standards)codes based on average workforce challenge levels are displayed in a bar chart in Figure 16. Following counties like 20015 and 20105 (≈ 4.5), county 20077 (≈ 5.0) has the highest rating, suggesting extremely significant reported workforce problems. The remaining counties cluster around 4.0, indicating that workforce difficulty is consistently high in these areas. This trend shows that workforce issues are common throughout several counties with just minor variations in severity rather than being isolated. This data supports the need for tailored policy and resource allocation in the project's context by highlighting regional variations in workforce-related concerns and confirming the conclusion that workforce challenges are a recurring concern across jurisdictions.

Regression Analysis:

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Separate Regression Interpretation

ARISE
- Workforce challenge: coef = -0.067, p = 0.233, not statistically significant
- Infrastructure stress: coef = -0.325, p = 0.000, statistically significant
- R-squared: 0.085

CivicPulse
- Workforce challenge: coef = -0.223, p = 0.002, statistically significant
- Infrastructure stress: coef = 0.610, p = 0.000, statistically significant
- R-squared: 0.371

Combined Regression Interpretation

const: coef = 4.475, p = 0.000, statistically significant
workforce_challenge_level: coef = -0.199, p = 0.000, statistically significant
infra_finance_stress_index: coef = -0.005, p = 0.938, not statistically significant
CivicPulse: coef = -0.298, p = 0.004, statistically significant
Combined R-squared: 0.059
```

Figure 17

The regression findings for the individual datasets (ARISE and CivicPulse) as well as the combined dataset are shown in Figure 17. Workforce challenge is not statistically significant in the ARISE model (coef ≈ -0.067, $p = 0.233$), whereas infrastructure stress has a low explanatory power ($R^2 \approx 0.085$) and is significantly and adversely correlated with fiscal situation (coef ≈ -0.325, $p < 0.001$). The CivicPulse model, on the other hand, indicates that both variables are statistically significant, with infrastructure stress positively associated (coef ≈ 0.610, $p < 0.001$) and workforce challenge negatively associated (coef ≈ -0.223, $p = 0.002$), as well as a higher R^2 (≈ 0.371). In the combined model, infrastructure stress becomes minor (coef ≈ -0.005, $p = 0.938$), workforce challenge is still significant (coef ≈ -0.199, $p < 0.001$), and the dataset source (CivicPulse) is significant (coef ≈ -0.298, $p = 0.004$), with a low overall fit ($R^2 \approx 0.059$), indicating limited explanatory power of the combined model.

These findings support the project's conclusion that these connections are dataset-dependent and should be read cautiously since they show that relationships between variables differ between datasets and are typically weak or inconsistent.

RESULTS

Several recurring patterns pertaining to local government capability are shown by the study of the harmonized dataset. The bulk of replies fall within the moderate-to-high range (levels 3 and 4), indicating the widespread prevalence of workforce difficulties across jurisdictions. Although the distributions from ARISE and CivicPulse are comparable, ARISE often reports somewhat higher workforce difficulty levels. Bar charts, histograms, and boxplots all consistently display this pattern, suggesting that workforce-related constraints are enduring throughout local governments.

On the other hand, fiscal conditions seem to be consistent among datasets. The majority of replies (about 3 to 4) are in the mid-to-high range, indicating that local governments generally view their financial situation as relatively stable. Boxplots and bar charts are examples of visualizations that demonstrate no difference between ARISE and CivicPulse, supporting the uniformity of financial situations across sources.

On the other hand, infrastructure finance stress exhibits much more fluctuation. The distributions exhibit a broad range of values, including both high and low extremes, suggesting that infrastructure-related financial stress varies across jurisdiction. Boxplots, which show wider distributions and the existence of outliers in both datasets, provide more evidence for this variability. Regression analysis offers more information about how

variables relate to one another. Infrastructure stress has a strong negative correlation with fiscal situation, although workforce difficulty is not statistically significant in the ARISE dataset. Although they have different consequences, infrastructure stress and workforce issues are both statistically significant in the CivicPulse dataset.

Workforce problems are still statistically significant in the combined model, but infrastructure stress becomes insignificant, and the overall explanatory power is still modest ($R^2 = 0.059$), suggesting that these factors by themselves are insufficient to explain fiscal consequences.

Furthermore, grant capacity varies by jurisdiction; the CivicPulse dataset shows greater average values than ARISE. Indicators of interorganizational difficulties and governance show moderate distributions, which reflect variations in institutional capacity among local governments. Overall, the results indicate that infrastructure stress is more changeable and context-dependent, whereas workforce difficulties are permanent and regularly influential. Although it is often weak, there is a little negative correlation between workforce difficulties and the state of the economy. These findings show how intricate local government systems are and how crucial it is to use harmonized data for accurate and insightful analysis. We also performed supplementary analysis in Power BI.

RECOMMENDATIONS

Several suggestions to enhance comprehension and decision-making concerning local:-

1. government capability can be made in light of the findings: Make workforce development a top priority: Workforce issues have a long-lasting effect on local

government capability and are regularly noted across datasets. Operational efficiency can be increased by addressing staffing shortages, enhancing workforce planning, and funding skill development.

2. Adopt focused infrastructure tactics: Policymakers should use region-specific remedies rather than standard methods due to the substantial variety in infrastructure funding stress. To properly understand these differences, additional data sources including funding levels, demographics, and geographical features should be included.
3. Boost grant capacity support: CivicPulse's higher grant capacity indicates variations in funding availability or competence. Initiatives for resource sharing, technical assistance, and training can enhance grant procurement across jurisdictions.
4. Enhance data standardization: To simplify harmonization and increase comparability, future data gathering initiatives should concentrate on uniform variable definitions and answer formats.
5. Make use of multi-method analytical techniques: For a more thorough understanding, decision-making should incorporate descriptive, visual, and statistical techniques because regression models exhibit limited explanatory ability.
6. Extend future studies: By incorporating more factors and larger datasets, differences in institutional ability and fiscal position can be better explained.

CONCLUSION

This study successfully illustrates the process of harmonizing two separate survey datasets into a single analytical framework. A consistent dataset of 19 harmonized variables was produced for analysis through crosswalk development, codebook interpretation, and data standardization. The results show that while personnel issues continue to be a common and enduring problem, local governments often enjoy moderate fiscal stability.

On the other hand, disparate financial strains are shown by the significant differences in infrastructure finance stress between jurisdictions. The intricacy of local government systems is shown by regression analysis, which also shows that connections between important variables are typically weak and dataset-dependent.

All things considered, this study emphasizes how crucial data harmonization is to facilitating insightful cross-dataset comparisons. Additionally, it emphasizes the importance of integrating regression-based, exploratory, and descriptive approaches to better comprehend real-world problems and facilitate data-driven, well-informed decision-making.

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APPENDICES

A: Data Sources and Description

To understand variable definitions, survey questions, and response structures, this study utilized two independently collected survey datasets along with their corresponding codebooks.

ARISE Kansas City Survey

- 309 observations
- Includes variables related to infrastructure systems, governance, fiscal condition, and hazard preparedness
- Contains Likert-scale, ranking-based, and binary variables

ARISE Codebook

- Provides detailed descriptions of survey questions and variables
- Explains response options and coding schemes
- Helps interpret different types of variables (ordinal, binary, ranking)
- Used to understand how responses are structured and recorded

CivicPulse Kansas Sample

- 135 observations
- Focuses on workforce challenges, fiscal condition, and infrastructure-related issues
- Uses different naming conventions and coding structures

CivicPulse Codebook

- Contains definitions of survey questions and variables

- Describes response categories and measurement scales
- Helps interpret differences in coding and structure compared to ARISE
- Used to ensure correct understanding before harmonization

B: Data Harmonization Process

Among the steps in the harmonization process were:

- Using codebooks to find overlapping constructs
- Making a crosswalk table to map variables across datasets
- Standardizing the names and forms of variables
- Transforming answers into a standard scale (1–5, if applicable)
- Handling missing values (such as NA and "Don't know")
- Combining datasets and including a source identifier
- Consistency and comparability between datasets were guaranteed by this procedure

C: Key Harmonized Variables

```
Final harmonized variables (present in both datasets):
```

	harmonized_variable	arise_column	civic_column
0	workforce_challenge_level	q8	workforce_sc
1	fiscal_condition_overall	q119	condition_sc
2	service_early_warning_available	q9_early_warning	services_gr_early
3	service_backup_electric_available	q9_backup_electric	services_gr_backup
4	service_evacuation_plan_available	q9_evacuation_route_or_plan	services_gr_evac
5	service_utility_shutoff_assistance_available	q9_financial_assistance_for_low_income_shut_offs	services_gr_financial
6	service_heating_cooling_stations_available	q9_heating_or_cooling_stations	services_gr_heating
7	early_warning_multilingual	q9_early_warning_lang	early_sc
8	evacuation_support_no_car_households	q9_evacuation_plan	evacuation_sc
9	equity_group_elderly_support	q10_elderly_people	groups_gr_elderly
10	equity_group_low_income_support	q10_low_income	groups_gr_low
11	equity_group_homeless_support	q10_homeless	groups_gr_homeless
12	equity_group_non_english_support	q10_non_english	groups_gr_non
13	respondent_tenure_local_gov	q29	tenure_sc
14	respondent_education_level	q30	education_sc
15	infra_finance_spending_gap	q13_reduce_services	infra_gr_spending
16	infra_finance_unstable_funding	q13_reduce_fund_balance	infra_gr_relies
17	infra_finance_budget_constraints	q13_defer_capital_projects	infra_gr_budget
18	grant_capacity_success	q5	infra_gr_competes

```
Total common harmonized variables: 19
```

The final dataset included 19 key variables, such as:

1. workforce_challenge_level
2. fiscal_condition_overall
3. service_early_warning_available
4. service_backup_electric_available
5. service_evacuation_plan_available
6. service_utility_shutoff_assistance_available
7. service_heating_cooling_stations_available

8. early_warning_multilingual
9. evacuation_support_no_car_households
10. equity_group_elderly_support
11. equity_group_low_income_support
12. equity_group_homeless_support
13. equity_group_non_english_support
14. respondent_tenure_local_gov
15. respondent_education_level
16. infra_finance_spending_gap
17. infra_finance_unstable_funding
18. infra_finance_budget_constraints
19. grant_capacity_success

These factors, which include personnel, financial health, infrastructural stress, and readiness, are important aspects of local government capacity.

D: Analytical Methods

The following methods were applied:

- Descriptive statistics
- Distribution analysis (histograms)
- Comparative analysis (bar charts, boxplots)
- Cross-tabulation analysis
- Correlation analysis
- Regression analysis (separate and combined models)

These methods helped examine relationships between workforce, fiscal, and infrastructure factors.

E: Visualizations

The project includes the following visualizations:

- Bar charts comparing dataset averages
- Histograms showing distributions of key variables
- Boxplots for variability comparison
- Stacked bar charts for categorical comparisons
- Line Chart for relationship analysis
- Government type distribution charts
- Workforce and fiscal condition by government type
- Respondent role comparison charts
- Governance practice index analysis
- Grant capacity analysis
- Interorganizational difficulty analysis

These visualizations support interpretation and enhance analytical insights.

F: Dashboard

The dashboard provides an overview of workforce challenges, fiscal condition, and infrastructure stress across the harmonized ARISE and CivicPulse datasets.

A Power BI dashboard was created to visually summarize the harmonized ARISE and CivicPulse datasets. The dashboard presents key indicators related to workforce challenges, fiscal condition, infrastructure stress, and dataset comparisons. Overall, the dashboard shows that most local governments report moderate workforce challenges and average to good fiscal condition, with generally weak relationships between variables.